26900058 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCBTG300,

Discipline Specific Elective (DSE)- MG5DSEBBT304

RECOMBINANT DNA TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One Word Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Name any two antibiotics that belongs to the class of aminoglycosides. [U] / [CO2]
2. State the purpose of using adapters in molecular cloning. [U] / [CO1]
3. Mention two applications of RAPD. [U] / [CO3]
4. What is the tet regulatory system? [U] / [CO2]
5. What is meant by primer extension? [U] / [CO3]
6. Name the vector used for gene transfer in both bacteria and plants. [K] / [CO1]
7. Define In-Vivo gene therapy. [K] / [CO3]
8. Which gene is used in blue-white screening? [U] / [CO2]

9. Name any two qualitative agronomic traits improved through rDNA technology. [K] / [CO3]

10. What information can be obtained from a footprinting experiment? [An] / [CO3]

11. Name the process of assembling bacteriophage particles outside a living cell. [U] / [CO1]

12. What is the major difference between electroporation and microinjection ?. [K] / [CO2]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Explain the significance of the intergenic region and cloning site in M 13 vector construction. [U] / [CO1]

14. Discuss about the Hygromycin and Puromycin resistance in animals. [K] / [CO2]

15. Write a short note on recombinant insulin. [U] / [CO3]

16. Describe the characteristic features of pSC vectors. [U] / [CO1]

17. Discuss the importance of affinity tags in downstream processing of recombinant proteins. [U] / [CO2]

18. How do restriction endonucleases facilitate DNA manipulation in genetic engineering? [U] / [CO1]

[5x4 = 20]

Part C

Essay Questions

Answer any **1** questions

Each question carries **10** marks

19. Describe the organization of a typical expression cassette and explain the functions of its major components from promoter to transcription termination. [U] / [CO2]

20. Discuss the major NGS platforms—Illumina, SOLiD, Ion Torrent and 454. [K] / [CO3]

[10x1 = 10]